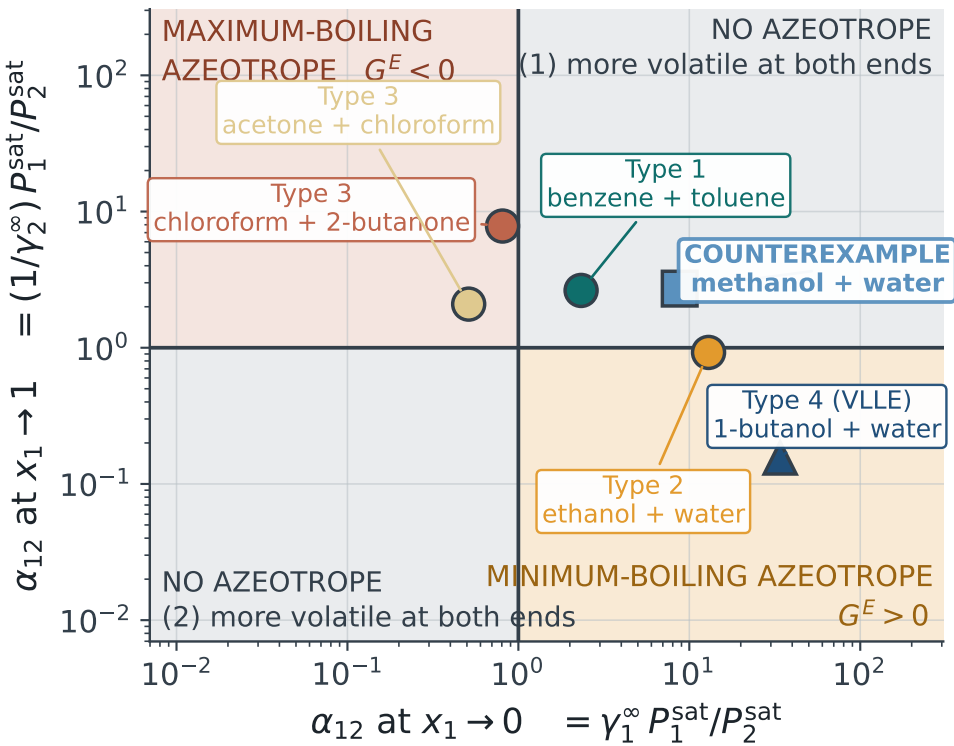
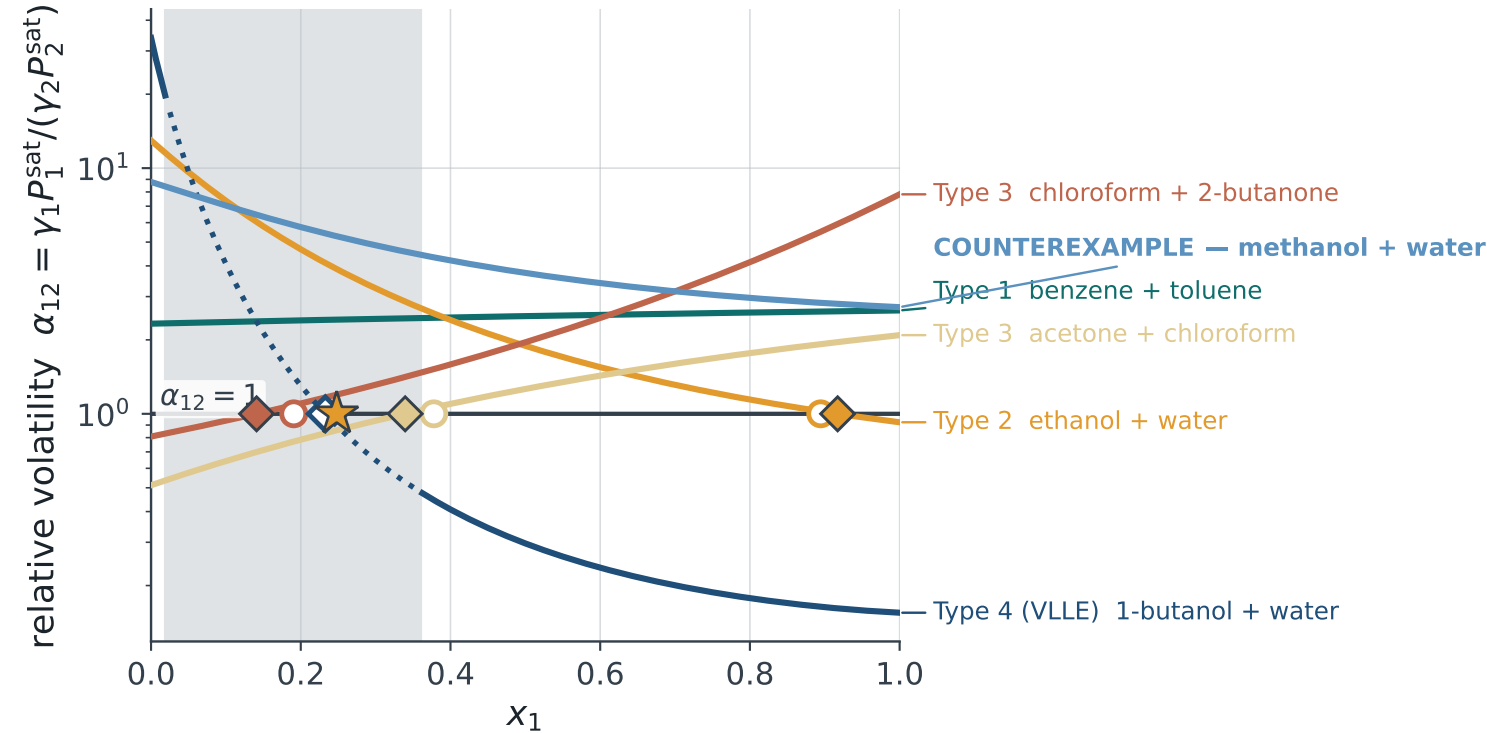


An azeotrope is a consequence, not a category: $\gamma_1/\gamma_2 = P_2^{\text{sat}}/P_1^{\text{sat}}$ either has a root in (0, 1) or it does not

(a) the criterion as a map: an azeotrope exists iff α_{12} straddles 1



(b) the same statement in composition space



filled diamond = azeotrope of the fitted NRTL; open circle = crossing of $y_1 = x_1$ read from the published table
 ethanol + water: model 0.917 vs data 0.895; chloroform + 2-butanone: model 0.141 vs data 0.191; acetone + chloroform: model 0.339 vs data 0.378
 shaded strip = the MEASURED two-liquid region of 1-butanol + water; the dotted curve inside it is model, not measurement.
 Open diamond = the model's root, which falls inside that region; star = the measured heteroazeotropic vapour, $y_1 = 0.248$.

γ_i^{∞} from NRTL ($\alpha = 0.3$) fitted to each published set by Barker's method; P_i^{sat} from the Antoine constants in vlekkit.data.LIBRARY, evaluated at each end's own boiling temperature for the isobaric sets. Straddling ends guarantee a root; non-straddling ends do not forbid one — a G^E with three or more parameters can make γ_1/γ_2 non-monotonic and place two azeotropes with both ends on the same side. WHAT THIS MAP DOES NOT DECIDE. It decides azeotropy. It does not decide how many liquid phases there are. 1-butanol + water lands in the same quadrant as ethanol + water because both have a minimum-boiling azeotrope; that its azeotrope is HETEROGENEOUS, and that its liquid splits in two, is a stability question this figure cannot answer. Module 5 answers it.